D03 · Industrial Automation & Robotics

02 · Collaborative robots (cobots) — when and how

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1 WHAT IT IS

Collaborative robots (cobots) are robotic arms designed to operate safely alongside humans without physical separation — built with force-limiting joints, rounded geometry, and ISO/TS 15066-compliant safety functions that permit shared workspace operation. The defining feature is not the robot itself but the work architecture: a cobot deployed in collaboration with an operator captures hybrid value that neither pure manual nor pure industrial-robot deployment can produce; a cobot deployed in isolation behind a fence is an industrial robot that costs more for less performance.

2 WHY IT MATTERS

The cobot category emerged from a specific gap in industrial automation — the tasks that are too low-volume, too variable, or too ergonomically problematic for an industrial robot, but too repetitive or physically taxing for sustained manual operation. **The dominant failure mode is treating cobots as cheap industrial robots — deploying them behind fences, removing the operator, optimizing for cycle time. This captures the lower cost but eliminates the entire reason cobots exist**. The technical specification that defines collaborative operation is explicit on this point **(ISO/TS 15066:2016 — international technical specification, "Robots and robotic devices — Collaborative robots")**: collaborative operation is defined as four specific modes (safety-rated monitored stop, hand guiding, speed and separation monitoring, power and force limiting), and cobot value emerges from the work-architecture choice between these modes, not from the robot's mere presence. Industry deployment data shows the asymmetry: **cobots deployed in genuinely collaborative operation generate 2–4x the ROI of cobots deployed as cheap industrial robots**, because the collaboration unlocks tasks that pure automation cannot perform economically.

3 HOW IT IS DONE

1. **Choose cobot over industrial robot when the case requires human judgment, variable handling, or ergonomic relief — not when capex is constrained.** The right cobot cases are tasks with one repetitive sub-task (cobot does this) and one variable or judgment-driven sub-task (operator does this). Pick-and-place inspection-and-packaging, machine tending with operator-driven changeover, assembly with operator-driven quality check. If the case is fully deterministic and high-volume, the industrial robot is correct; the cobot is the wrong tool because the collaboration adds no value.
2. **Design the work architecture explicitly — which sub-tasks the cobot does, which the operator does, where the hand-off happens.** Cobot value lives in the hand-off. The design must specify: cobot fetches and presents the part, operator inspects and decides, cobot packs the accepted part, operator handles the rejected part. The hand-off geometry, the timing, and the cycle interaction determine whether the architecture works ergonomically and economically. Skipping this design produces cobots that operate alongside operators in parallel rather than in collaboration.
3. **Run a risk assessment specific to ISO/TS 15066 collaborative operation — separate from the ISO 10218 industrial-robot assessment.** Collaborative operation has its own safety envelope: contact forces and pressures must be within biomechanical limits for the body region exposed, transient and quasi-static loading must be characterized, the contact zones must be identified through risk assessment, not assumed. Plants that deploy cobots without a TS 15066 risk assessment either remove the operator (defeating the purpose) or expose the operator to undocumented contact risk.
4. **Program the cobot using the right method for the task — teach pendant, hand-guiding, simulation, or code.** Most cobot value comes from rapid reprogramming for changeovers; the programming method must match operator capability. Hand-guiding (operator physically moves the cobot through the motion, robot records) works well for simple pick-and-place; teach pendant for moderate complexity; simulation/offline programming for complex multi-step tasks. Forcing the wrong programming method against the task collapses the deployment velocity that cobots are supposed to enable.
5. **Train operators as cobot co-workers, not as cobot supervisors — the work relationship is symmetric, not hierarchical.** Operator training must cover: when to override the cobot, how to reprogram for changeovers, how to recognize cobot misbehavior, how to perform daily checks. Operators who treat the cobot as a supervised machine produce parts; operators trained as co-workers produce continuous improvement ideas that compound the deployment's value over time.

6. **Track ergonomic outcomes alongside throughput — the cobot's secondary ROI lives in injury reduction and operator retention.** Cobot deployments typically pay back primary capex on throughput and quality; the secondary ROI (reduced repetitive strain injuries, lower workers' comp claims, improved operator retention in physically demanding work) is frequently larger than the primary ROI but rarely measured. Plants that track ergonomic outcomes alongside production outcomes capture the full economic case; plants that track only throughput leave 30–50% of the value invisible.

WORKED EXAMPLE — ILLUSTRATIVE

A CPG manufacturer runs a manual case-packing operation on 4 lines, packing variable-mix retail cases with 6–14 SKUs per case. Baseline: **2 operators per line, throughput 18 cases/hour, repetitive-strain-injury (RSI) claims averaging 3.4/year/line ($82K average claim cost), case-mix variability prevents traditional industrial-robot deployment**. The cobot case-match: high task variability (cobots win), ergonomic burden (cobots win), need for operator judgment on case mix (cobot + operator wins). The implementation: **1 UR10e cobot per line, deployed in shared workspace with 1 operator** — cobot fetches cartons from line buffer and orients them, operator hand-packs the SKU mix (judgment task), cobot closes and palletizes the completed case. ISO/TS 15066 risk assessment defines the contact-force envelope; hand-guiding programming for changeover; operators trained as co-workers including reprogramming authority. Results at 14 months: **throughput 18 -> 22 cases/hour (+24%), labor restructured to 1.2 operator/line average (from 2), RSI claims drop from 3.4/year/line to 0.7/year/line (-79%), payback 14 months on $98K cobot + EOAT investment**. The secondary ROI from injury reduction ($232K/year across 4 lines) exceeds the primary throughput ROI; the operators who would have rotated out of the role in 2–3 years now stay, preserving institutional knowledge that the throughput numbers do not capture.

IN PRACTICE

Choose cobot over industrial robot only when the task requires human judgment or ergonomic relief

Cobots emerged for tasks too low-volume or variable for industrial robots but too repetitive for sustained manual operation. Pure-substitution applications — fence the cobot, remove the operator — should use an industrial robot at lower cost and higher cycle time.

Design the work architecture explicitly; the hand-off is the engineering object

What does the operator do, what does the cobot do, where is the hand-off, in which of the four ISO/TS 15066 collaborative-operation modes **(ISO/TS 15066:2016 — international technical specification)**. Skip this design and the cobot is an expensive substitute for either pure automation or pure manual.

Operations leads, with safety as constraint and engineering as integrator

Safety-driven deployment converges toward isolation (operator removed, fence added) to minimize regulatory exposure; engineering-driven deployment converges toward maximum cycle time (operator removed, cobot at full speed). Both eliminate the collaboration that makes deployment economic.

Track ergonomic outcomes alongside throughput

Cobot deployments unlock value on the long tail of ergonomically problematic tasks — value invisible in throughput-only metrics. RULA/REBA scores, MSD incident rates, operator retention on the cell — these are the leading indicators that the work architecture is real.

WHAT YOU WOULD USE

COLLABORATIVE ROBOT PLATFORMS

Universal Robots (UR3e / UR5e / UR10e / UR16e / UR20) — The dominant collaborative-robot OEM with the largest installed base and strongest ecosystem of EOAT, programming tools, and integration partners — the default starting point for most cobot deployments **(Universal Robots — vendor)**.

FANUC CRX / ABB GoFa & SWIFTI / Yaskawa HC series — Cobot lines from the established industrial-robot OEMs — strongest when the operation already has installed base of the same vendor's industrial robots and wants ecosystem continuity **(FANUC — vendor; ABB — vendor; Yaskawa — vendor)**.

Doosan / Techman / Aubo / Productive Robotics — Mid-tier cobot platforms with competitive specifications at lower price points; strong fit for operations testing cobots before committing to enterprise-scale deployment, or for very price-sensitive cases **(Doosan Robotics — vendor; Techman — vendor)**.

Cobot + EOAT marketplace (OnRobot, Robotiq, Schunk) + in-house deployment — For operations with sufficient engineering capability — cobot from any vendor, EOAT from the open marketplace, deployment and integration in-house. The path with the lowest TCO and highest

flexibility; requires the operations engineering muscle to be real.

4 SO WHAT

The cobot deployed behind a fence, with the operator removed, is an expensive industrial robot — the value of cobots lives in the work architecture, not in the price point. Choose cobot over industrial robot when the task requires human judgment or ergonomic relief, design the work architecture explicitly with the hand-off as the engineering object, run an ISO/TS 15066 risk assessment specific to collaborative operation, match programming method to operator capability, train operators as co-workers not as supervisors, and track ergonomic outcomes alongside throughput. The 2–4x ROI gap between collaboratively-deployed cobots and isolated-deployment cobots is structural, not technical — and it lives entirely in whether the work architecture is designed or skipped.

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